HOT SPOTS ON SYMMETRIC CONVEX BODIES OF REVOLUTION

ABSTRACT. We give direct variational proofs of the strict hot-spots property for explicit classes of smooth convex bodies of revolution in every dimension $n \geq 2$. For a body symmetric across its equatorial plane, the first eigenspace consists of the lowest axial-odd mode, the lowest transverse-odd modes, or their direct sum. We prove strict directional monotonicity for each of the two types whenever it belongs to the first eigenspace. Consequently every first eigenfunction has strict hot spots whenever the axial and transverse eigenvalues are distinct. We provide explicit second-moment, long-body, and flat-body conditions which ensure this separation. The proof uses a convex tangent-vector variational principle, with its equality case proved in full, and sign changes of the meridional components of a gradient. No monotonicity theorem is assumed. Smooth positively curved examples, including nonellipsoidal quartic bodies, satisfy $d(\Omega)^n/|\Omega| \to \infty$. Arbitrary mixtures at an axial–transverse eigenvalue coincidence are outside the general hot-spots conclusion of this paper, except for the ball, which is treated separately.

Draft status. This candidate proof was obtained by our team on October 18, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction and main results

For a bounded connected smooth domain $\Omega \subset \mathbb{R}^n$, number the Neumann eigenvalues, with multiplicities, by

$$0 = \mu_0(\Omega) < \mu_1(\Omega) \le \mu_2(\Omega) \le \cdots .$$

The first eigenspace in this paper always means the eigenspace $E_1(\Omega)$ for the first *positive* eigenvalue μ_1. All eigenfunctions are real valued. We call the following the strict hot-spots property:

$$\min_{\partial\Omega} u < u(x) < \max_{\partial\Omega} u \qquad (x \in \Omega, \quad 0 \ne u \in E_1(\Omega)). \tag{1.1}$$

A strictly positive derivative in one fixed direction implies this property and excludes every interior critical point.

1.1. **Research context.** Jerison and Nadirashvili proved the hot-spots property for convex planar domains with two axes of symmetry, including multiple first positive eigenvalues [3]. Chen, Li, and Wang proved monotonicity of the least axially odd eigenfunction on suitable rotationally symmetric domains, and obtained hot-spots conclusions for long domains [1]. Their axial-odd eigenvalue need not be the first positive Neumann eigenvalue, so spectral selection is an essential part of a hot-spots argument. Kennedy and Rohleder developed a vector variational approach to higher-dimensional lip domains and related symmetric domains [4]. Mean-gradient arguments for convex symmetric bodies also appear in Klartag [5, Section 4].

Convexity alone is insufficient in arbitrary dimension: de Dios Pont constructed convex counterexamples in all sufficiently large dimensions [2]. The present argument uses rotational symmetry to reduce the relevant boundary tangency relation to two meridional components. The vector variational inequality and its equality case are proved here, as are the monotonicity and spectral-selection steps. The cited rotational monotonicity theorem is background rather than an input to these proofs.

1.2. **The geometric class and the spectral alternatives.** Write $x = (y, z) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and $\rho = |y|$. We consider

$$\Omega = \{(y, z) : -L < z < L, \ |y| < r(z)\}, \qquad R = r(0) > 0, \tag{1.2}$$

under the following hypotheses:

(G1) r is even, positive and C^∞ on $(-L, L)$, extends continuously to $[-L, L]$, and $r(\pm L) = 0$;

(G2) r is concave and $r'(z) < 0$ for $0 < z < L$;

(G3) the resulting boundary $\partial\Omega$ is C^∞.

These conditions describe smooth convex bodies with strictly decreasing upper meridian profile. Every smooth body of revolution with positive principal curvatures and the stated equatorial symmetry satisfies them. Smoothness is a condition on the hypersurface: the graph r need not be smooth at $\pm L$. In particular, the usual square-root behavior at a curved pole is allowed.

Let $P_\pm = (0, \pm L)$. Define the two odd-sector eigenvalues by

$$\lambda_\parallel = \inf_{\substack{0 \neq f \in H^1(\Omega) \\ f(y,-z) = -f(y,z)}} \frac{\int_\Omega |\nabla f|^2}{\int_\Omega f^2}, \tag{1.3}$$

$$\lambda_\perp = \inf_{\substack{0 \neq f \in H^1(\Omega) \\ f(-y_1,y_2,\ldots,z) = -f(y_1,y_2,\ldots,z)}} \frac{\int_\Omega |\nabla f|^2}{\int_\Omega f^2}. \tag{1.4}$$

Transverse rotations make the second definition independent of the chosen transverse direction. The symbol λ_{odd} will also denote $\lambda_\parallel$ in the geometric estimates.

Theorem 1.1 (Spectral alternatives and monotonicity)**.** *For a body satisfying* (G1)–(G3)*,*

$$\mu_1(\Omega) = \min\{\lambda_\parallel, \lambda_\perp\}.$$

The following conclusions hold.

(i) *If $\lambda_\parallel < \lambda_\perp$, the first eigenvalue is simple. Its eigenfunction U, chosen positive on $z > 0$, is invariant under transverse rotations, odd in z, and satisfies*

$$U_z > 0 \quad \text{in } \Omega, \qquad U_\rho < 0 \quad \text{when } z > 0,\ \rho > 0.$$

Its unique maximum and minimum are P_+ and P_-.

(ii) *If $\lambda_\perp < \lambda_\parallel$, the first eigenvalue has multiplicity $n-1$. For every nonzero first eigenfunction, a choice of unit vector $e \in \mathbb{R}^{n-1}$ gives*

$$\partial_e u > 0 \quad \text{in } \Omega.$$

Its unique maximum and minimum are $(Re, 0)$ and $(-Re, 0)$.

(iii) *If $\lambda_\parallel = \lambda_\perp$, the multiplicity is n. The axial subspace and the transverse subspace separately have the monotonicity and extremum conclusions in* (i) *and* (ii)*. No conclusion for an arbitrary mixture of these two subspaces is included in this assertion.*

In particular, if $\lambda_\parallel \neq \lambda_\perp$, every first eigenfunction satisfies (1.1) *and has no interior critical point.*

The distinction at a spectral coincidence is substantive. Monotonicity of separate eigenfunctions does not imply monotonicity of their linear combinations. All geometric hypotheses in the next theorem give a strict spectral separation, so that issue does not arise there.

For a ball, its additional rotational symmetries handle every mixture; Corollary 6.1 gives a proof within the same framework. More generally, fix any profile satisfying (G1)–(G3) and stretch the body in the axial direction by a factor $t > 0$. Proposition 6.2 shows that only finitely many values of t can be spectral coincidences. Thus the full hot-spots conclusion holds throughout every such stretching family apart from a finite set of possible exceptions of precisely this spectral type.

1.3. **Explicit geometric conditions.** Set $\beta_m^2 = \mu_1(B^m)$ for the unit m-ball, with $\beta_1 = \pi/2$, and $\alpha_n^2 = \lambda_1^D(B^n)$. Thus $\alpha_n = j_{n/2-1,1}$. Define

$$M_\parallel = \frac{1}{|\Omega|} \int_\Omega z^2 \, \mathrm{d}x, \qquad M_\perp = \frac{1}{|\Omega|} \int_\Omega y_1^2 \, \mathrm{d}x.$$

Theorem 1.2 (Moment and aspect-ratio conditions)**.** *For a body satisfying* (G1)–(G3)*, either condition*

$$M_\parallel \geq \frac{R^2}{\beta_{n-1}^2} \qquad \text{or} \qquad \frac{L}{R} \geq \frac{\alpha_n}{\beta_{n-1}} \tag{1.5}$$

implies the axial conclusion of Theorem 1.1. A sufficient condition involving only elementary constants is

$$\frac{L}{R} \geq \sqrt{\frac{(n+1)(n+2)}{2n}}. \tag{1.6}$$

In dimensions three and four, $L \geq 2R$ is therefore sufficient.

Either condition

$$M_\perp \geq \frac{4L^2}{\pi^2} \qquad \text{or} \qquad \frac{L}{R} \leq \frac{\pi}{2}\sqrt{\frac{n}{(n+1)(n+2)}} \tag{1.7}$$

implies the transverse conclusion of Theorem 1.1. All displayed nonstrict inequalities are permitted, including their endpoints.

Finally, the geometric range is unbounded in the diameter–volume quantity requested here. With $\omega_m = |B^m|$, define

$$\Omega_{L,R,c} = \left\{(y,z) : \frac{|y|^2}{R^2} + (1-c)\frac{z^2}{L^2} + c\frac{z^4}{L^4} < 1\right\}, \qquad 0 < c < 1. \tag{1.8}$$

These are smooth positively curved nonellipsoidal bodies. Section 6 proves that they satisfy the axial theorem for all sufficiently large L/R, and that, for $L/R \geq \sqrt{1+c}$,

$$\frac{d(\Omega_{L,R,c})^n}{|\Omega_{L,R,c}|} = \frac{2^n}{\omega_{n-1} I_{n,c}} \left(\frac{L}{R}\right)^{n-1} \longrightarrow \infty, \qquad I_{n,c} = \int_{-1}^{1} [(1-s^2)(1+cs^2)]^{(n-1)/2}\, \mathrm{d}s. \tag{1.9}$$

Here $d(\Omega)$ is the Euclidean diameter. Thus no fixed upper bound on $d(\Omega)^n/|\Omega|$ is required by these results.

1.4. **Outline of the proof.** We first prove the convex tangent-vector inequality $\mathcal{Q}(X) \geq \mu_1 \|X\|_2^2$ and show that equality consists exactly of gradients of first Neumann eigenfunctions. The divergence-free part of a tangent field has zero mean in every Cartesian component; this is the key observation in the proof. Convexity and a patch of positive curvature supplied by an enclosing sphere make the equality case rigid. The resulting injective mean-gradient map reduces the first eigenspace to the two odd-sector alternatives above.

For an axial first eigenfunction, a sign change of its radial derivative and an absolute value of its axial derivative preserve the vector energy and boundary tangency. The equality characterization then forces the desired signs. For a transverse mode, an analogous change lowers an explicit angular energy term, and equality again determines the signs. The constructions are justified in H^1 across the axis and equatorial plane; no regularity of polar coordinates at those sets is assumed.

The spectral estimates then follow from transverse slice Poincaré inequalities, linear trial functions, and a comparison with a double-cone weight. The final section computes volumes and diameters of the example families and explains why axial stretching produces arbitrarily large diameter–volume ratios within the proved class.

2. A vector variational principle on a smooth convex domain

Throughout this section, $\Omega \subset \mathbb{R}^n$ is bounded and convex with C^∞ boundary. Write $\mu = \mu_1(\Omega) > 0$ for its first positive Neumann eigenvalue, and set

$$E_1(\Omega) = \{u \in H^1(\Omega) : -\Delta u = \mu u,\ \partial_\nu u = 0\}.$$

The equations in this definition mean that

$$\int_\Omega \nabla u \cdot \nabla \varphi = \mu \int_\Omega u\varphi \qquad \text{for every } \varphi \in H^1(\Omega).$$

Taking $\varphi = 1$ gives zero mean, and elliptic regularity makes these solutions smooth up to the boundary. We use the outward second fundamental form

$$\mathrm{II}_x(S,T) = \langle D_S\nu, T\rangle, \qquad S,T \in T_x\partial\Omega.$$

Thus II is nonnegative on a convex domain. The scalar Poincaré inequality is

$$\int_\Omega |\nabla f|^2 \geq \mu \int_\Omega |f - f_\Omega|^2, \qquad f_\Omega = \frac{1}{|\Omega|}\int_\Omega f. \tag{2.1}$$

For a mean-zero f, equality holds precisely when $f \in E_1(\Omega)$, including the zero function. To see the forward implication, take any mean-zero $\varphi \in H^1(\Omega)$ and apply the inequality to $f + t\varphi$. At equality the resulting nonnegative quadratic polynomial has a minimum at $t = 0$, so $\int_\Omega \nabla f \cdot \nabla\varphi = \mu \int_\Omega f\varphi$. Replacing an arbitrary test function by $\varphi - \varphi_\Omega$ extends this identity to all of $H^1(\Omega)$. Conversely, testing the weak eigenvalue equation with f proves equality.

Let

$$\mathcal{X}(\Omega) = \{X \in H^1(\Omega;\mathbb{R}^n) : X \cdot \nu = 0 \text{ on } \partial\Omega \text{ in the trace sense}\}.$$

For $X \in \mathcal{X}(\Omega)$, define

$$\mathcal{Q}(X) = \int_\Omega \left((\operatorname{div} X)^2 + \sum_{1\leq i<j\leq n} (\partial_i X_j - \partial_j X_i)^2 \right). \tag{2.2}$$

The displayed sum fixes the normalization of the antisymmetric derivative in every dimension.

Lemma 2.1 (Smooth approximation with tangent boundary values)**.** *The smooth vector fields on $\overline{\Omega}$ which are tangent to $\partial\Omega$ are dense in $\mathcal{X}(\Omega)$ in the H^1 norm. Consequently, for every $X \in \mathcal{X}(\Omega)$,*

$$\mathcal{Q}(X) = \int_\Omega |DX|^2 + \int_{\partial\Omega} \mathrm{II}(X,X). \tag{2.3}$$

Proof. Take a finite smooth partition of unity on $\overline{\Omega}$ subordinate to interior charts and boundary charts, with each localization supported away from the edges of its chart. On each boundary chart choose a smooth orthonormal frame consisting, on the boundary, of $n-1$ tangent vectors and the normal vector; extend the frame smoothly into a collar. After localization, the normal coefficient of X has zero boundary trace. Its extension by zero outside the chart therefore belongs to $H^1_0(\Omega)$. It can be approximated in H^1 by smooth coefficients vanishing near the boundary, using the density of $C^\infty_c(\Omega)$ in $H^1_0(\Omega)$. The tangential coefficients can be approximated by arbitrary smooth coefficients up to the boundary. Multiplication by a further cutoff, equal to one on the support of the localized field, keeps all these approximations inside the chosen chart. Multiplication by this fixed cutoff and by the smooth frame is continuous in H^1. Recombining the coefficients therefore produces smooth tangent fields converging to the localized field in H^1. Interior smoothing and the partition of unity finish the approximation. The usual scalar approximation up to a smooth boundary used here follows, for example, by a local extension across the flattened boundary and convolution.

For a smooth field, expansion of the squares gives

$$(\operatorname{div} X)^2 + \sum_{i<j}(\partial_i X_j - \partial_j X_i)^2 - |DX|^2 = \operatorname{div}\big(X \operatorname{div} X - (X\cdot\nabla)X\big).$$

On the boundary the first term in this flux vanishes because $X\cdot\nu = 0$. Differentiating that equality in the tangent direction X gives

$$-\langle (X\cdot\nabla)X, \nu\rangle = \langle X, D_X\nu\rangle = \mathrm{II}(X,X).$$

The divergence theorem proves (2.3) for smooth tangent fields. Pass to the just-constructed approximations. The bulk terms are continuous in H^1, and the boundary term is continuous under the L^2 convergence of traces because II is bounded. This proves the identity for every $X \in \mathcal{X}(\Omega)$. □

Lemma 2.2 (A patch of positive curvature)**.** *There is a nonempty relatively open patch $\Gamma \subset \partial\Omega$ on which* II *is positive definite. Moreover, if a Neumann eigenfunction vanishes on such a patch, then it vanishes identically. In particular, if $Z \in \mathcal{X}(\Omega)$ has every Cartesian component in $E_1(\Omega)$ and $\int_{\partial\Omega}\mathrm{II}(Z,Z) = 0$, then $Z = 0$.*

Proof. Choose $c \in \Omega$, and let p maximize $|x-c|$ on $\overline{\Omega}$. Set $\rho = |p-c| > 0$. The maximum lies on the boundary, since an interior maximum would have $p = c$. Tangential stationarity gives $p - c = a\nu(p)$. The supporting half-space and $c \in \Omega$ give $(c-p)\cdot\nu(p) < 0$, so $a = \rho$. For $T \in T_p\partial\Omega$, choose a smooth boundary curve γ with $\gamma(0) = p$ and $\gamma'(0) = T$. Differentiating $\gamma'\cdot\nu(\gamma) = 0$ gives $\gamma''(0)\cdot\nu(p) = -\mathrm{II}_p(T,T)$. The second derivative of $|\gamma - c|^2$ at its maximum is therefore

$$0 \geq 2|T|^2 - 2\rho\,\mathrm{II}_p(T,T).$$

It follows that $\mathrm{II}_p \geq \rho^{-1}I$, and continuity gives the required patch.

For the second assertion, let $-\Delta f = \lambda f$ be a Neumann eigenfunction vanishing on an open boundary patch. Both f and $\partial_\nu f$ are zero there. Choose a ball B centered at a point of the patch with $B \cap \partial\Omega$ contained in that patch, and let F equal f on $B\cap\Omega$ and zero on $B\setminus\overline{\Omega}$. For every $\varphi \in C_c^\infty(B)$, Green's formula gives

$$\int_B F(-\Delta\varphi - \lambda\varphi) = \int_{B\cap\partial\Omega}(\varphi\partial_\nu f - f\partial_\nu\varphi) = 0.$$

There are no terms on ∂B because the test function is compactly supported in B. Thus F solves the constant-coefficient Helmholtz equation distributionally throughout B. Interior elliptic regularity and analyticity make it a real analytic solution, so its vanishing on the exterior open part forces vanishing throughout B. Analytic continuation in the connected interior of Ω then gives $f = 0$ everywhere. The continuation argument needs only an open smooth boundary patch; its positive curvature is not used in this last step.

Finally, under the hypotheses on Z, its components are smooth up to the boundary. The continuous function $\mathrm{II}(Z,Z)$ is nonnegative and has integral zero, so it vanishes on the boundary. Positive definiteness on Γ and tangency give $Z = 0$ there. Applying the boundary continuation just proved to each component gives $Z = 0$ throughout Ω. □

Lemma 2.3 (The tangent-vector ground state)**.** *For every $X \in \mathcal{X}(\Omega)$,*

$$\mathcal{Q}(X) \geq \mu\int_\Omega |X|^2. \tag{2.4}$$

Equality holds if and only if $X = \nabla p$ for some $p \in E_1(\Omega)$.

Proof. The zero normal trace gives $\int_\Omega \operatorname{div} X = 0$ by the H^1 Green formula. On the mean-zero subspace of $H^1(\Omega)$, the Poincaré inequality makes the Dirichlet form coercive. The Lax–Milgram theorem therefore gives a unique mean-zero $p \in H^1(\Omega)$ satisfying

$$\int_\Omega \nabla p \cdot \nabla \varphi = \int_\Omega X \cdot \nabla \varphi = -\int_\Omega (\operatorname{div} X)\varphi.$$

The equality with the last integral again uses the zero normal trace of X. Compatibility extends the identity to every $\varphi \in H^1(\Omega)$. Since $\operatorname{div} X \in L^2(\Omega)$, Neumann regularity on the smooth domain gives $p \in H^2(\Omega)$, with $\Delta p = \operatorname{div} X$ and $\partial_\nu p = 0$. Put $W = X - \nabla p$. Then $W \in \mathcal{X}(\Omega)$ and $\operatorname{div} W = 0$. The weak construction also gives

$$\int_\Omega W \cdot \nabla \varphi = 0 \qquad (\varphi \in H^1(\Omega)). \tag{2.5}$$

Taking $\varphi = p$ proves

$$\int_\Omega W \cdot \nabla p = 0, \qquad \|X\|_{L^2}^2 = \|\nabla p\|_{L^2}^2 + \|W\|_{L^2}^2.$$

The divergence and antisymmetric derivatives also separate:

$$\mathcal{Q}(X) = \|\Delta p\|_{L^2}^2 + \mathcal{Q}(W). \tag{2.6}$$

The gradient part satisfies

$$\|\nabla p\|_{L^2}^2 = -\int_\Omega p \Delta p \le \|p\|_{L^2}\|\Delta p\|_{L^2} \le \mu^{-1/2}\|\nabla p\|_{L^2}\|\Delta p\|_{L^2}.$$

Consequently

$$\|\Delta p\|_{L^2}^2 \ge \mu \|\nabla p\|_{L^2}^2. \tag{2.7}$$

Suppose equality holds in (2.7). If $\nabla p = 0$, its zero mean gives $p = 0$. Otherwise the preceding chain has equal endpoints and forces equality in (2.1). Thus $p \in E_1(\Omega)$ in either case.

The key point for the remaining field is that every Cartesian component of W has zero mean. Take $\varphi = x_i$ in (2.5) to obtain

$$\int_\Omega W_i = \int_\Omega W \cdot \nabla x_i = 0. \tag{2.8}$$

Apply the scalar Poincaré inequality to each W_i and use convexity in (2.3). This yields

$$\mathcal{Q}(W) = \int_\Omega |DW|^2 + \int_{\partial\Omega} \mathrm{II}(W, W) \ge \mu \int_\Omega |W|^2. \tag{2.9}$$

Combining (2.6)– (2.9) proves (2.4).

Suppose equality holds. Both inequalities (2.7) and (2.9) must then be equalities. In the latter, all componentwise Poincaré inequalities are equalities and

$$\int_{\partial\Omega} \mathrm{II}(W, W) = 0.$$

Thus every W_i belongs to $E_1(\Omega)$. The final assertion of Lemma 2.2 applies to the tangent field W and gives $W = 0$. Equality in the gradient inequality then gives $X = \nabla p$ with $p \in E_1(\Omega)$.

Conversely, if $p \in E_1(\Omega)$, its gradient has tangent trace and zero antisymmetric derivative, and

$$\mathcal{Q}(\nabla p) = \int_\Omega (\Delta p)^2 = \mu^2 \int_\Omega p^2 = \mu \int_\Omega |\nabla p|^2.$$

This proves the equality characterization. □

Lemma 2.4 (Injectivity of the mean gradient)**.** *The linear map*

$$J : E_1(\Omega) \longrightarrow \mathbb{R}^n, \qquad J(u) = \int_\Omega \nabla u$$

is injective. More precisely,

$$\frac{\mu}{|\Omega|}|J(u)|^2 \geq \int_{\partial\Omega} \mathrm{II}(\nabla_\tau u, \nabla_\tau u) \qquad (u \in E_1(\Omega)). \tag{2.10}$$

Proof. Apply (2.3) to the tangent field ∇u. The last calculation in Lemma 2.3 gives

$$\int_\Omega |D^2 u|^2 + \int_{\partial\Omega} \mathrm{II}(\nabla_\tau u, \nabla_\tau u) = \mu \int_\Omega |\nabla u|^2.$$

On the other hand, applying (2.1) to each $\partial_i u$ and adding gives

$$\int_\Omega |D^2 u|^2 \geq \mu \int_\Omega |\nabla u|^2 - \frac{\mu}{|\Omega|}|J(u)|^2.$$

The two formulas prove (2.10).

If $J(u) = 0$, each $\partial_i u$ has zero mean. The preceding two formulas and convexity force the curvature integral to vanish and every componentwise Poincaré inequality to be an equality. Thus $\partial_i u \in E_1(\Omega)$ by the scalar equality characterization, including its Neumann boundary condition. The final assertion of Lemma 2.2, now applied to the tangent field ∇u, gives $\nabla u = 0$. The zero mean of u then gives $u = 0$, proving injectivity. □

Corollary 2.5 (Symmetry of the first eigenspace)**.** *One has* $\dim E_1(\Omega) \leq n$*. If an orthogonal map* S *preserves* Ω*, and* $(T_S u)(x) = u(S^{-1}x)$*, then*

$$J(T_S u) = S J(u). \tag{2.11}$$

If $\Omega = -\Omega$*, every first eigenfunction is centrally odd:*

$$u(-x) = -u(x). \tag{2.12}$$

Proof. The dimension bound follows from injectivity. Orthogonal changes of variables preserve the weak Neumann equation, so T_S maps $E_1(\Omega)$ to itself. Since $\nabla(T_S u)(x) = S\nabla u(S^{-1}x)$ and $|\det S| = 1$, integration gives (2.11). For $S = -I$, this yields $J(T_{-I}u) = -J(u) = J(-u)$. Injectivity gives $T_{-I}u = -u$, which is (2.12). □

3. The first eigenspace and the axial variational proof

Assume (G1)–(G3) throughout this section, and let $d = n - 1$. On the nonpolar part of the curved boundary, an outward unit normal is

$$\nu = \frac{e_\rho - r'(z)e_z}{\sqrt{1 + r'(z)^2}}. \tag{3.1}$$

In particular $\nu_\rho > 0$, and $z\nu_z > 0$ when $z \neq 0$. The Neumann condition for a rotational function is therefore

$$U_\rho = r'(z)U_z. \tag{3.2}$$

3.1. **Odd ground states and spectral classification.**

Lemma 3.1 (Coordinate-odd ground states)**.** *The lowest eigenvalue in each coordinate-odd subspace is simple in that subspace. Its eigenfunction can be chosen positive on the corresponding positive half-domain, is even in every other principal coordinate, and is smooth on* $\overline{\Omega}$*.*

Proof. Restriction to the positive half-domain identifies the odd variational problem with a mixed problem: Dirichlet data on the central flat section and Neumann data on the exterior boundary. More precisely, its form domain consists of the H^1 functions on that half-domain whose trace vanishes on the flat section; odd extension across that section is an H^1 function on Ω and multiplies both quadratic integrals by two; the matching zero traces prevent any jump across the reflection plane. This extension has zero mean, so (2.1) bounds its quotient below by $\mu_1 > 0$. The half-domain is bounded, convex and connected, hence Lipschitz. Compactness of its H^1 embedding in L^2 and weak closedness of the zero-trace condition give a normalized minimizer. Replacing a minimizer by its absolute value preserves its quotient; interior elliptic regularity and the strong minimum principle make this minimizer strictly positive in the connected interior.

Every lowest eigenfunction has one sign: otherwise its absolute value, also a smooth positive interior minimizer, would vanish at an interior point. Two linearly independent lowest eigenfunctions would have a nonzero linear combination orthogonal to a positive one, contradicting this sign property. This proves simplicity.

The odd extension is a weak Neumann eigenfunction on the full smooth domain. To check this, split any test function into its odd and even parts; the odd part restricts to an admissible mixed test, and the even part has zero pairing by parity. Full-domain Neumann regularity therefore gives smoothness up to the boundary, including the intersection of the reflection plane with the curved boundary. Reflections in all other coordinates preserve the positive normalized mixed ground state, so simplicity makes it even in those coordinates. □

Let U be the positive-on-$z > 0$ axial ground state. Every orthogonal transformation of y preserves its mixed problem and its positive L^2 normalization. Simplicity therefore gives $U = U(\rho, z)$; when $d = 1$, the transverse reflection supplies the same conclusion. Let u_i, $1 \le i \le d$, be the transverse ground states, positive on $y_i > 0$. Their eigenvalues are all $\lambda_\perp$.

Proposition 3.2 (The axial and transverse alternatives)**.** *One has* $\mu_1 = \min\{\lambda_\parallel, \lambda_\perp\}$*, and*

$$E_1(\Omega) = \begin{cases} \operatorname{span}\{U\}, & \lambda_\parallel < \lambda_\perp, \\ \operatorname{span}\{u_1, \dots, u_d\}, & \lambda_\perp < \lambda_\parallel, \\ \operatorname{span}\{U, u_1, \dots, u_d\}, & \lambda_\parallel = \lambda_\perp. \end{cases} \tag{3.3}$$

Whenever the transverse modes belong to $E_1(\Omega)$*, they can be normalized so that*

$$(u_1(y,z), \dots, u_d(y,z)) = h(\rho^2, z)y, \qquad h(\rho^2, z) > 0 \quad (0 < \rho < r(z)), \tag{3.4}$$

where h *is smooth at the axis and even in* z*.*

Proof. Coordinate reflections act on $E_1(\Omega)$ as commuting self-adjoint involutions. A joint parity component f can have $\int_\Omega \partial_i f \neq 0$ only if it is odd in coordinate i and even in all the others. All other parity patterns have zero mean gradient and hence vanish by Lemma 2.4. Every surviving component attains the lowest Rayleigh quotient in its odd subspace: its quotient is μ_1, whereas every odd function has mean zero and quotient at least μ_1. Lemma 3.1 therefore identifies it with its odd ground state. At least one such component is nonzero, while $\mu_1 \le \lambda_\parallel$ and $\mu_1 \le \lambda_\perp$ by the same mean-zero variational principle. This proves the minimum formula. Every ground state whose odd-sector eigenvalue equals μ_1 is itself in $E_1(\Omega)$, so the parity decomposition proves exactly the direct sum (3.3).

Suppose the transverse modes are first modes. Parity gives $J(u_i) = c_i e_i$. Testing their full-domain weak eigenvalue equation with $y_i \in H^1(\Omega)$ gives

$$c_i = \int_\Omega \partial_i u_i = \lambda_\perp \int_\Omega y_i u_i > 0.$$

The last inequality follows from the chosen sign of u_i on the positive half-domain and its oddness. Positive rescaling therefore normalizes the modes so that $J(u_i) = e_i$.

For $S \in O(d)$ acting on y and fixing z, equivariance and injectivity of J give $T_S u_i = \sum_j S_{ji} u_j$, since both sides have mean gradient Se_i. For $\Phi = (u_1, \dots, u_d)$ this is exactly $\Phi(Sy, z) = S\Phi(y, z)$. The subgroup fixing a nonzero y then forces $\Phi(y, z)$ to be parallel to y, and rotations make its coefficient a function only of ρ and z. When $d = 1$, the same conclusion follows from oddness in y_1.

To justify smoothness also at boundary points of the axis, take a smooth extension $\widetilde{\Phi}$ to a neighborhood of $\overline{\Omega}$ invariant under transverse orthogonal maps and $z \mapsto -z$. Replace it by

$$\widehat{\Phi}(y, z) = \frac{1}{2} \int_{O(d)} S^{-1} \big(\widetilde{\Phi}(Sy, z) + \widetilde{\Phi}(Sy, -z) \big)\, dS,$$

where dS is normalized Haar measure. The transformation law just proved and evenness in z show that this average agrees with Φ on $\overline{\Omega}$; it is smooth, equivariant, and even in z. Set $g(t, z) = \widehat{\Phi}_1(te_1, z)$. It is smooth and odd in t, so

$$\frac{g(t, z)}{t} = \int_0^1 \partial_t g(\tau t, z)\, d\tau$$

extends smoothly as an even function of t. Its Taylor expansion through degree $2N$ has only even powers, and the remainder after k derivatives in $s = t^2$ is $O(s^{N+1-k})$, by the chain rule and Taylor's differentiated remainder estimates. The same estimates hold after any fixed number of z derivatives. Since N is arbitrary, substitution $t = \sqrt{s}$ gives smooth one-sided derivatives of every order at $s = 0$, jointly with z. This gives the required smooth $h(\rho^2, z)$, including at the poles; equivariance reconstructs the full vector. The same even-function argument applies to smooth rotationally invariant scalar functions. Finally, positivity of $u_1(\rho e_1, z)$ for $0 < \rho < r(z)$ gives the stated interior positivity of h, and evenness in z is preserved throughout. □

3.2. **Changing the signs of a meridional gradient.**

Proposition 3.3 (Axial monotonicity)**.** *If $\lambda_\parallel = \mu_1(\Omega)$, then its positive axial ground state satisfies*

$$U_z > 0 \quad \text{in } \Omega, \qquad U_\rho < 0 \quad \text{for } z > 0,\ \rho > 0. \tag{3.5}$$

Its unique maximum and minimum on $\overline{\Omega}$ are P_+ and P_-. Moreover, $U_z > 0$ on $\partial\Omega \setminus \{P_+, P_-\}$. The assertion holds for the axial mode even if the transverse eigenvalue is equal to it.

Proof. At $\rho > 0$ put $p = U_z$ and $q = U_\rho$, and define the vector field

$$X = |p| e_z - \operatorname{sgn}(z) |q| e_\rho. \tag{3.6}$$

We first justify its regularity across the sets where these coordinates or signs are singular. By the smooth even-function argument in Proposition 3.2, write $U(\rho, z) = H(\rho^2, z)$ with H smooth and odd in z. Smooth division of $2H_s$ by z gives

$$q(\rho, z) = \rho z K(\rho^2, z)$$

with K smooth, including at the axis and equatorial plane. The transverse part of (3.6) is therefore exactly $-z|K(|y|^2, z)|y$. Together with $X_z = |U_z|$, this defines a locally Lipschitz field near $\overline{\Omega}$, and in particular an H^1 field. A smooth symmetric extension of U justifies the same statement at the poles and the boundary.

The Neumann relation (3.2), and the sign of r' on each half, show on the nonpolar boundary that

$$X = \operatorname{sgn}(p)\nabla U$$

where $p \neq 0$; both fields are zero there if $p = 0$. The same identity for the quadratic boundary quantities holds at the equator and poles by continuity. Thus $X \cdot \nu = 0$ and $\mathrm{II}(X, X) = \mathrm{II}(\nabla U, \nabla U)$ on the boundary.

For a meridional field $A(\rho, z)e_\rho + B(\rho, z)e_z$, direct differentiation in an orthonormal polar frame gives

$$|DX|^2 = A_\rho^2 + A_z^2 + B_\rho^2 + B_z^2 + (n-2)\frac{A^2}{\rho^2}.$$

Applied to (3.6), the chain rule for absolute values shows, almost everywhere, that $|DX|^2 = |D^2U|^2$ and $|X|^2 = |\nabla U|^2$. The H^1 justification above ensures that no distribution supported on the axis or equatorial plane is omitted. The Gaffney identity now gives

$$\mathcal{Q}(X) = \mathcal{Q}(\nabla U) = \mu_1 \int_\Omega |\nabla U|^2 = \mu_1 \int_\Omega |X|^2.$$

By Lemma 2.3, $X = \nabla V$ for some $V \in E_1(\Omega)$. The field X is equivariant under every transverse orthogonal transformation S. Consequently $\nabla(V \circ S) = S^T X(Sx) = X(x) = \nabla V$. The difference $V \circ S - V$ is constant on the connected domain, and its zero mean makes that constant zero. Thus V is invariant under all these transformations, including the reflection when $d = 1$. Proposition 3.2 makes that invariant subspace exactly $\operatorname{span}\{U\}$. Hence $X = c\nabla U$, and equality of norms gives $c = 1$ or $c = -1$. The latter would make $U_z \leq 0$ everywhere, contrary to $U(0, 0) = 0$ and $U(0, z) > 0$ for $0 < z < L$. Therefore $c = 1$, proving the nonstrict signs in (3.5).

The nonzero function $p = U_z \geq 0$ satisfies $-\Delta p = \mu_1 p \geq 0$. The strong minimum principle yields $p > 0$ in the entire domain. On the connected meridian region $0 < \rho < r(z)$, $0 < z < L$, the function $w = -U_\rho \geq 0$ satisfies

$$w_{\rho\rho} + \frac{n-2}{\rho} w_\rho + w_{zz} + \left(\mu_1 - \frac{n-2}{\rho^2}\right) w = 0.$$

On any sufficiently small neighborhood away from $\rho = 0$, its coefficients are bounded. Choose a constant $C \leq \min\{0, \inf(\mu_1 - (n-2)/\rho^2)\}$ on that neighborhood. Then

$$w_{\rho\rho} + \frac{n-2}{\rho} w_\rho + w_{zz} + Cw \leq 0.$$

The strong minimum principle, applied locally and then using connectedness, makes w either positive throughout the meridian region or identically zero. The latter would make U independent of ρ on the upper half; the Neumann condition and $r'(z) < 0$ would then give $U_z(z) = 0$ for every $0 < z < L$, contradicting the already proved positivity of p. Thus $w > 0$. The same local supersolution inequality also permits Hopf's boundary point lemma at every nonpolar upper boundary point.

We also prove the asserted boundary positivity of $p = U_z$. At a nonpolar upper boundary point, $\nu_\rho, \nu_z > 0$. If $p = 0$ there, the Neumann relation gives $q = 0$, hence $\nabla U = 0$. For every tangent vector T, differentiating the Neumann condition at this critical point gives $0 = D^2U(T, \nu) + \nabla U \cdot D_T\nu = D^2U(T, \nu)$. Symmetry of the Hessian therefore gives

$$D^2U\,\nu = A\nu, \qquad A = U_{\nu\nu}.$$

Hopf's boundary point lemma applied to $p > 0$ yields $\partial_\nu p = A\nu_z < 0$, and hence $A < 0$. Applied in the smooth meridian region to $w = -U_\rho > 0$, it yields $\partial_\nu w = -A\nu_\rho < 0$, and hence $A > 0$, a contradiction. At an equatorial boundary point, $q = 0$ by oddness of U and $\nu_z = 0$. A hypothetical zero of p would again imply $\nabla U = 0$ and then $\partial_\nu p = A\nu_z = 0$, contradicting Hopf's lemma for p. Reflection gives the lower half. At the two poles $p = 0$ by the Neumann condition, as expected.

To locate the extrema, first take a point in the closed upper half-domain with $\rho > 0$. Strict radial decrease, applied first on an interior subsegment and then by continuity to a possible boundary endpoint, gives $U(\rho, z) < U(0, z)$ when $z > 0$. Strict axial increase similarly gives $U(0, z) < U(0, L)$ for $0 \le z < L$. The equatorial values are zero and the values on the negative half are nonpositive. Hence the unique maximum is P_+; oddness proves the unique minimum at P_-. □

3.3. **A transverse slice barrier.**

Theorem 3.4 (Axial spectral selection)**.** *If*

$$\lambda_{\parallel} < \frac{\beta_{n-1}^2}{R^2}, \tag{3.7}$$

then $\lambda_{\parallel} < \lambda_{\perp}$, and the full first eigenspace has all the axial conclusions of Theorem 1.1.

Proof. For any y_1-odd function $f \in H^1(\Omega)$, almost every transverse slice has zero mean on $B_{r(z)}^{n-1}$. Slice Poincaré and Fubini give

$$\int_\Omega |\nabla f|^2 \ge \int_{-L}^{L} \frac{\beta_{n-1}^2}{r(z)^2} \int_{B_{r(z)}^{n-1}} f(y,z)^2 \, \mathrm{d}y \, \mathrm{d}z \ge \frac{\beta_{n-1}^2}{R^2} \int_\Omega f^2.$$

For precision, first apply the slice inequality to smooth approximants on the truncated domains $\Omega \cap \{|z| < L - \varepsilon\}$. Antisymmetrization preserves the coordinate-odd condition, and the weight $r(z)^{-2}$ is bounded on each such domain. Thus H^1 convergence passes the inequality to f there. Letting $\varepsilon \downarrow 0$ by monotone convergence justifies the display on the whole domain and proves finiteness of its weighted integral, despite the singular weight at the poles. In fact the bound for the lowest transverse eigenvalue is strict:

$$\lambda_{\perp} > \frac{\beta_{n-1}^2}{R^2}.$$

To see this, apply the displayed inequality to its nonzero ground state, whose existence was proved in Lemma 3.1. The strict decrease of r on $(0, L)$ and its evenness give $r(z) < R$ for every $z \neq 0$. A nonzero L^2 function has positive slice mass on a set of z of positive measure, so the second inequality in the display is strict for that ground state. Attainment is used here: strict inequalities for individual trial functions alone would not imply a strict bound on their infimum. Combine this with (3.7), Proposition 3.2, and Proposition 3.3. □

4. The transverse first modes and flat bodies

Use the notation of Proposition 3.2, with $x = (y, z) \in \mathbb{R}^d \times \mathbb{R}$, $d = n - 1$, $\rho = |y|$, and $R = r(0)$. The eigenvalues $\lambda_{\parallel}$ and $\lambda_{\perp}$ are the lowest eigenvalues in the axial-odd and a transverse-coordinate-odd subspace, respectively. In particular, $\mu_1(\Omega) = \min\{\lambda_{\parallel}, \lambda_{\perp}\}$. Assume (G1)–(G3) throughout this section.

The upper meridional boundary can also be written as $z = f(\rho)$. Here f is the inverse of r restricted to $(0, L)$, extended by $f(0) = L$ and $f(R) = 0$. The inverse function theorem gives, on $(0, R)$,

$$f'(\rho) = \frac{1}{r'(f(\rho))} < 0, \qquad f''(\rho) = -\frac{r''(f(\rho))}{r'(f(\rho))^3} \le 0.$$

Thus f is decreasing and concave in every dimension under consideration.

Proposition 4.1 (Strict monotonicity of a transverse first mode)**.** *Suppose $\lambda_\perp = \mu_1(\Omega) = \mu$. For a unit vector $e \in \mathbb{R}^d$, let u_e be the corresponding transverse mode, chosen positive when $y \cdot e > 0$. Then*

$$\partial_e u_e > 0 \quad \text{in } \Omega \quad \text{and on } \partial\Omega \setminus \{(Re, 0), (-Re, 0)\}. \tag{4.1}$$

Its unique maximum and minimum are attained at $(Re, 0)$ and $(-Re, 0)$, respectively. Writing

$$u_e(y, z) = v(\rho, z)\frac{y \cdot e}{\rho}, \qquad v(\rho, z) > 0 \quad (0 < \rho < r(z)),$$

one has

$$v_\rho > 0 \quad \text{in the meridional interior with } \rho > 0, \qquad v_z < 0 \quad \text{there when } z > 0. \tag{4.2}$$

These assertions apply to each transverse mode even if $\lambda_\perp = \lambda_\parallel$; they do not assert a conclusion for a mixture of the two branches at such a tie.

Proof. A transverse orthogonal change of coordinates reduces the proof to $e = e_1$ in every dimension, including $d = 1$. First assume $d \geq 2$, and put

$$Y(\theta) = \theta_1, \qquad e_\rho = \theta = \frac{y}{\rho}, \qquad a = v_\rho, \quad b = \frac{v}{\rho}, \quad c = v_z.$$

Proposition 3.2 gives

$$v(\rho, z) = \rho h(\rho^2, z), \qquad h(\rho^2, -z) = h(\rho^2, z). \tag{4.3}$$

The same proposition supplies smoothness of h up to the axis and the boundary, including the poles, and its evenness in z. In particular, $b = h > 0$ in the interior when $\rho > 0$, and $b \geq 0$ on the closure by continuity. The gradient is

$$\nabla u = aYe_\rho + b\nabla_S Y + cYe_z, \qquad \nabla_S Y = e_1 - Ye_\rho. \tag{4.4}$$

Define a vector field by

$$X = |a|Ye_\rho + b\nabla_S Y - \operatorname{sgn}(z)|c|Ye_z. \tag{4.5}$$

We first justify its regularity. Its transverse component is

$$X_y = be_1 + (|a| - b)Ye_\rho.$$

Equation (4.3) gives $a - b = 2\rho^2 h_s(\rho^2, z)$, where s is the first variable of h. Since $b \geq 0$, the reverse triangle inequality yields

$$\big||a| - b\big| \leq |a - b| = O(\rho^2).$$

Thus the apparently singular angular term has a continuous extension through the axis. Its angular derivatives divided by ρ are $O(\rho)$. Moreover $a_\rho, b_\rho = O(\rho)$, so the Lipschitz chain rule makes its radial derivative $O(\rho)$; its axial derivative is bounded. These estimates are uniform locally up to the boundary. There is no extra weak derivative supported on the axis: inserting a radial cutoff at distance ε from it in the distributional derivative calculation gives an error $O(\varepsilon^{d-1})$, which tends to zero for every $d \geq 2$. The axial component has the nonsingular expression

$$X_z = -\operatorname{sgn}(z)|h_z(\rho^2, z)|y_1.$$

Smooth division by z gives $h_z(s, z) = zk(s, z)$ with k smooth. Thus $X_z = -z|k(\rho^2, z)|y_1$ is locally Lipschitz across $z = 0$ and the axis. A finite cover of $\overline{\Omega}$, using the smooth extensions already described at the poles, now gives $X \in H^1(\Omega; \mathbb{R}^n)$.

The three summands in (4.4) are orthogonal pointwise, so

$$|X|^2 = |\nabla u|^2. \tag{4.6}$$

For completeness, with $\nabla_m = (\partial_\rho, \partial_z)$, the Cartesian derivative energy of a field of the form on the right of (4.4) is

$$|D(\nabla u)|^2 = Y^2(|\nabla_m a|^2 + |\nabla_m c|^2) + (1-Y^2)|\nabla_m b|^2 + \frac{(a-b)^2(1+(d-2)Y^2) + c^2(1-Y^2)}{\rho^2}. \tag{4.7}$$

To verify the angular part, write the transverse field as $be_1 + (a-b)Ye_\rho$ and differentiate in an orthonormal tangent frame on S^{d-1}. For a unit tangent vector τ,

$$D_\tau(Ye_\rho) = (e_1 \cdot \tau)e_\rho + Y\tau.$$

The two summands are orthogonal. Summing their squared norms over the frame gives $(1-Y^2)+(d-1)Y^2 = 1+(d-2)Y^2$. The axial field contributes $c^2|\nabla_S Y|^2 = c^2(1-Y^2)$. The meridional part follows directly from the orthogonality of Ye_ρ, $\nabla_S Y$, and Ye_z. Replacing a, c as in (4.5) preserves the squared meridional weak derivatives almost everywhere and preserves c^2. Since $b > 0$, it gives

$$|D(\nabla u)|^2 - |DX|^2 = \frac{4b(-a)_+(1+(d-2)Y^2)}{\rho^2} \geq 0 \quad \text{almost everywhere.} \tag{4.8}$$

On the curved boundary the normal is meridional. Write its components as (ν_ρ, ν_z); away from the poles, $\nu_\rho > 0$, and ν_z has the sign of z. Applying the Neumann condition at the representative point $(\rho e_1, z)$, where $Y = 1$, gives the meridional identity

$$\nu_\rho a + \nu_z c = 0. \tag{4.9}$$

For $z \neq 0$, the replacement of (a, c) in (4.5) therefore multiplies both entries by the same sign, or leaves both zero. At $z = 0$ both entries vanish: $a = 0$ by (4.9) and $c = 0$ by reflection. The azimuthal component $b\nabla_S Y$ is unchanged. Hence X is tangent to the boundary. On a surface of revolution the meridional tangent line and the azimuthal tangent space are orthogonal principal spaces of the second fundamental form: differentiating (3.1) in a meridional tangent direction T keeps $D_T\nu$ in the meridional plane, whereas an azimuthal unit vector E satisfies $D_E\nu = (\nu_\rho/\rho)E$. Thus the second fundamental form has no mixed term between these spaces. The replacement changes only the sign of the meridional tangent component, and consequently

$$\mathrm{II}(X, X) = \mathrm{II}(\nabla u, \nabla u) \quad \text{almost everywhere on } \partial\Omega. \tag{4.10}$$

The two poles have surface measure zero, so no coordinate assertion at those points is required for this equality.

The Gaffney identity and (4.8)– (4.10) imply

$$\mathcal{Q}(X) \leq \mathcal{Q}(\nabla u) = \mu\|\nabla u\|_{L^2}^2 = \mu\|X\|_{L^2}^2.$$

Lemma 2.3 forces equality and gives $X = \nabla p$ for a first eigenfunction p. The nonnegative right-hand side of (4.8) must therefore have zero integral. Since $d \geq 2$, its angular factor is at least one, and $b > 0$ forces $a \geq 0$ almost everywhere. Continuity away from the axis upgrades this to every meridional point. Thus the transverse components of X and ∇u agree. Each transverse slice of Ω is a connected ball, so $p - u$ depends only on z. Its axial derivative is, for $\rho > 0$,

$$(-\operatorname{sgn}(z)|c| - c)Y.$$

A function independent of y cannot have this nonconstant angular dependence: evaluate at $\theta = e_1$ and $\theta = -e_1$ on each slice. The displayed coefficient must vanish. This proves $c \leq 0$ for $z > 0$, without any assumption about simplicity of the full first eigenspace.

Now

$$w := \partial_{y_1} u = aY^2 + b(1-Y^2) \geq 0.$$

Continuity gives the same nonnegativity on the axis. The function is nonzero, since $b > 0$, and satisfies $-\Delta w = \mu w \geq 0$ in Ω. The strong minimum principle therefore gives $w > 0$ throughout the interior. Evaluating at $\theta = e_1$ proves $a > 0$ for $\rho > 0$. Substitution of $u = vY$ in its eigenvalue equation, using $-\Delta_S Y = (d-1)Y$, gives the amplitude equation

$$v_{zz} + v_{\rho\rho} + \frac{d-1}{\rho} v_\rho + \left(\mu - \frac{d-1}{\rho^2}\right) v = 0.$$

Its coefficients are independent of z, so $-c = -v_z \geq 0$ solves the same equation in the connected upper meridional region. Away from $\rho = 0$, its coefficients are locally bounded. The local supersolution argument in the proof of Proposition 3.3 therefore applies: lower the zeroth-order coefficient to a nonpositive constant, and use the strong minimum principle. Thus $-c$ is either positive throughout that region or identically zero. In the latter case write $v(\rho, z) = V(\rho)$ there. Each $0 < \rho < R$ occurs on the upper cap, where (4.9) gives $V'(\rho) = 0$, contradicting the already proved $a > 0$. Hence $c < 0$ in the upper meridional interior. The same local supersolution argument also justifies the boundary point lemma for $-c$ below.

We give the boundary argument to rule out additional extrema. First, $v > 0$ on the meridional boundary wherever $\rho > 0$. Indeed, at the representative point $(\rho e_1, z)$ the eigenfunction is positive on the adjacent interior of the half-domain $y_1 > 0$; this point is away from its flat Dirichlet boundary. A zero boundary value there would contradict its Neumann condition by Hopf's boundary point lemma. Thus $b > 0$ at all these points.

Consider a point $(\rho e_1, z)$ on the boundary with $\rho, z > 0$. If $w = 0$ there, then $a = 0$ and (4.9) gives $c = 0$, so $\nabla u = 0$. The differentiated Neumann identity at a boundary critical point, established in the proof of Proposition 3.3, gives

$$D^2 u\, \nu = A\nu, \qquad A = u_{\nu\nu}.$$

Hopf's lemma for the positive function w gives $\partial_\nu w = A\nu_\rho < 0$, hence $A < 0$. Apply Hopf's lemma also to the positive meridional function $-c$. Its outward meridional normal is (ν_ρ, ν_z), and at the point in question

$$\partial_\nu(-c) = -A\nu_z < 0,$$

which gives $A > 0$, a contradiction. Reflections in z and y_1 give the same conclusion at the other non-equatorial points with $\theta = \pm e_1$.

At a boundary point with $\rho > 0$ and $\theta \neq \pm e_1$, the expression for w gives $w \geq b(1 - Y^2) > 0$. At an axial pole, (4.3) shows that all components of ∇u except possibly w vanish. If $w = 0$ there, the preceding differentiated Neumann identity again applies; since $\nu \cdot e_1 = 0$, it would give $\partial_\nu w = 0$, contradicting Hopf's lemma. We have proved (4.1). At $(\pm R e_1, 0)$ the direction e_1 is normal and $w = 0$ by the Neumann condition.

An interior extremum is impossible because $w > 0$. At any boundary extremum, tangential stationarity and the Neumann condition imply $\nabla u = 0$. The boundary positivity just proved leaves only $(\pm R e_1, 0)$. Their values have opposite signs, with the positive value at $(R e_1, 0)$, proving the asserted uniqueness.

When $n = 2$, exchange the two coordinates. The domain is again a smooth symmetric convex body of revolution, now with axis y_1. Its new profile is the even extension of the inverse function f defined at the start of this section, which is decreasing and concave on $(0, R)$. At the original upper pole the normal is e_z, so the smooth boundary writes z as a smooth even function of y_1 near zero. Thus this even extension is smooth at its equator, and the coordinate exchange preserves smoothness of the boundary. Together with the endpoint values of f, these facts show that the exchanged profile satisfies (G1)–(G3) as well. The mode under consideration is its axial-odd first mode and is even in its transverse coordinate. Proposition 3.3 therefore gives $\partial_{y_1} u > 0$ in the interior and on the boundary except at $(\pm R, 0)$, with these two points its unique extrema. For

$y_1 > 0$ its axial derivative in the exchanged coordinates is v_ρ, while its radial transverse derivative is v_z when $z > 0$. The two strict signs in (4.2) follow as well. □

4.1. **Explicit conditions selecting the transverse branch.** Define the transverse second moment, independent of the choice of unit $e \in \mathbb{R}^d$, by

$$M_\perp = \frac{1}{|\Omega|} \int_\Omega (y \cdot e)^2 \, dx.$$

Proposition 4.2 (Transverse moment criterion)**.** *If*

$$M_\perp \geq \frac{4L^2}{\pi^2}, \tag{4.11}$$

then $\lambda_\perp < \lambda_\parallel$. Every nonzero first eigenfunction is a transverse mode for a suitable direction and therefore satisfies Proposition 4.1. In particular, the first eigenvalue has multiplicity $n-1$.

Proof. The coordinate trial function $y \cdot e$ is odd in that coordinate and gives

$$\lambda_\perp < \frac{1}{M_\perp}.$$

The inequality is strict: equality would make this nonzero linear function an odd-sector ground state and hence, by Lemma 3.1, a full-domain Neumann eigenfunction with positive eigenvalue. Its zero Laplacian makes this impossible.

For almost every fixed y with $|y| < R$, an H^1 function w odd in z has a continuous odd representative on its vertical interval, so its trace at $z = 0$ is zero. Applying the interval Poincaré inequality on $(-f(|y|), f(|y|))$ and using oddness gives

$$\int_0^{f(|y|)} |\partial_z w|^2 \, dz \geq \frac{\pi^2}{4f(|y|)^2} \int_0^{f(|y|)} |w|^2 \, dz \geq \frac{\pi^2}{4L^2} \int_0^{f(|y|)} |w|^2 \, dz.$$

No condition at $z = f(|y|)$ is imposed in this inequality. For the Sobolev passage at the degenerating endpoints, use smooth odd approximants on $|y| < R - \varepsilon$, where $f(|y|)^{-2}$ is bounded, and then monotone convergence as in the proof of Theorem 3.4. Integration and odd reflection give $\lambda_\parallel \geq \pi^2/(4L^2)$. Consequently (4.11) implies $\lambda_\perp < \lambda_\parallel$. The first-eigenspace description and multiplicity follow from Proposition 3.2. □

Corollary 4.3 (An elementary flat-body criterion)**.** *The sufficient condition in Proposition 4.2 holds whenever*

$$\frac{L}{R} \leq \frac{\pi}{2} \sqrt{\frac{n}{(n+1)(n+2)}}. \tag{4.12}$$

Proof. The concavity and endpoint values of f established at the start of the section give the tangent-line inequality

$$f(\rho) + (R - \rho) f'(\rho) \geq 0,$$

so $q(\rho) = f(\rho)/(R-\rho)$ is nondecreasing on $(0, R)$. Relative to the probability measure proportional to $\rho^{d-1}(R - \rho) \, d\rho$, the functions $q(\rho)$ and ρ^2 therefore have nonnegative covariance. This follows directly by integrating $(q(s) - q(t))(s^2 - t^2) \geq 0$ against two copies of that measure. All the

required moments are finite, because multiplication by q replaces its density by $\rho^{d-1}f(\rho)$. The angular average of $(\theta\cdot e)^2$ is $1/d$, also for $d=1$. Polar integration therefore gives

$$\begin{aligned} M_\perp &= \frac{1}{d}\,\frac{\int_0^R \rho^{d+1}f(\rho)\,d\rho}{\int_0^R \rho^{d-1}f(\rho)\,d\rho} \\ &\geq \frac{1}{d}\,\frac{\int_0^R \rho^{d+1}(R-\rho)\,d\rho}{\int_0^R \rho^{d-1}(R-\rho)\,d\rho} = \frac{d+1}{(d+2)(d+3)}R^2 = \frac{n}{(n+1)(n+2)}R^2. \end{aligned}$$

Condition (4.12) therefore implies $M_\perp \geq 4L^2/\pi^2$, as required. □

5. Geometric conditions that select the axial eigenfunction

Assume (G1)–(G3) and retain the geometric and spectral notation of the preceding sections. In particular, $\lambda_{\mathrm{odd}} = \lambda_{\|}$ and $\beta_m^2 = \mu_1(B^m)$ for the unit m-ball, with $\beta_1 = \pi/2$. The purpose of this section is to give explicit sufficient conditions for

$$\lambda_{\mathrm{odd}} < \frac{\beta_{n-1}^2}{R^2}. \tag{5.1}$$

The spectral selection theorem, Theorem 3.4, then identifies this axial eigenvalue with the simple first positive Neumann eigenvalue of Ω.

5.1. **A lower bound for the transverse ball eigenvalue.**

Lemma 5.1. *For every integer $m \geq 1$,*

$$\beta_m^2 > m+1. \tag{5.2}$$

Proof. Project the unit sphere $S^{m+1} \subset \mathbb{R}^{m+2}$ onto its first m coordinates. Away from a set of surface measure zero, a parametrization is

$$(x, \sqrt{1-|x|^2}\cos\vartheta, \sqrt{1-|x|^2}\sin\vartheta), \qquad x \in B^m, \quad 0 \leq \vartheta < 2\pi.$$

The metric blocks in these coordinates are

$$g_{xx} = I + \frac{x\otimes x}{1-|x|^2}, \qquad g_{x\vartheta} = 0, \qquad g_{\vartheta\vartheta} = 1-|x|^2.$$

The x block has determinant $(1-|x|^2)^{-1}$ and inverse $I - x\otimes x$. Thus the full metric has determinant one, and the induced measure is $\mathrm{d}x\,\mathrm{d}\vartheta$. For smooth f on $\overline{B^m}$, its lift $\widetilde{f} = f(x)$ is smooth on the entire sphere: extend f smoothly past the closed ball and compose with the ambient coordinate projection. In particular, the collapsed circle fibers create no singularity in this function. The metric calculation gives

$$\int_{S^{m+1}} \widetilde{f}^2 = 2\pi\int_{B^m} f^2, \qquad \int_{S^{m+1}} |\nabla_S \widetilde{f}|^2 = 2\pi\int_{B^m} (|\nabla f|^2 - (x\cdot\nabla f)^2).$$

The measure identity also shows that a mean-zero f has a mean-zero lift. The spherical harmonic eigenvalues on S^{m+1} are $\ell(\ell+m)$, $\ell = 0, 1, \ldots$, as follows from the polar-coordinate formula for the Euclidean Laplacian on homogeneous harmonic polynomials. Their restrictions form a complete orthogonal decomposition of $L^2(S^{m+1})$: harmonic decomposition expresses every polynomial restriction as a sum of such harmonics, and polynomial restrictions are dense. If $\widetilde{f}_\ell$ is the degree-ℓ component of a smooth mean-zero lift, integration by parts and Parseval's identity therefore give

$$\int_{S^{m+1}} |\nabla_S \widetilde{f}|^2 = \sum_{\ell\geq 1} \ell(\ell+m)\|\widetilde{f}_\ell\|_2^2 \geq (m+1)\int_{S^{m+1}} \widetilde{f}^2.$$

Combining this gap with the two lift identities gives

$$\int_{B^m} |\nabla f|^2 \geq (m+1) \int_{B^m} f^2 + \int_{B^m} (x \cdot \nabla f)^2.$$

For a mean-zero $f \in H^1(B^m)$, choose smooth approximants and subtract their means. They converge in H^1, and all three ball integrals in the last inequality converge, since $|x| \leq 1$. This extends the inequality to every such f. Apply it to a nonzero first Neumann eigenfunction on the ball. Equality $\beta_m^2 = m+1$ would force $x \cdot \nabla f = 0$. Smoothness then makes f constant along every radial segment and continuity at the origin makes all these constants equal. This contradicts its nonzero, mean-zero normalization. The argument includes $m = 1$. □

5.2. **The axial second moment.**

Proposition 5.2 (Moment criterion)**.** *If*

$$\frac{1}{|\Omega|} \int_\Omega z^2 \, \mathrm{d}x \geq \frac{R^2}{\beta_{n-1}^2}, \tag{5.3}$$

then (5.1) *holds, including equality in* (5.3)*.*

Proof. The odd trial function $v(y, z) = z$ gives the strict bound

$$\lambda_{\text{odd}} < \frac{|\Omega|}{\int_\Omega z^2 \, \mathrm{d}x}. \tag{5.4}$$

Strictness is the coordinate-trial argument of Proposition 4.2, applied to the axial coordinate z instead of a transverse coordinate. Combining (5.4) and (5.3) proves the assertion. □

5.3. **Comparison with the double cone.** Let

$$\alpha_n^2 = \lambda_1^D(B^n), \qquad \alpha_n = j_{n/2-1,1}, \tag{5.5}$$

where λ_1^D is the first Dirichlet eigenvalue and $j_{\nu,1}$ is the first positive zero of the Bessel function J_ν. Only the variational definition and its radial ordinary differential equation are needed below.

Lemma 5.3 (Double-cone trial)**.** *For the domain under consideration,*

$$\lambda_{\text{odd}} < \frac{\alpha_n^2}{L^2}. \tag{5.6}$$

Proof. The compactness, absolute-value and simplicity arguments of Lemma 3.1, now applied to $H_0^1(B^n)$, give a simple positive Dirichlet ground state. Rotational invariance and simplicity make it radial. Dirichlet regularity makes its radial profile F smooth on $[0, 1]$; normalize it by $F(0) = 1$. It satisfies

$$F''(s) + \frac{n-1}{s} F'(s) + \alpha_n^2 F(s) = 0, \qquad F'(0) = 0, \qquad F(1) = 0.$$

For completeness, putting $\nu = n/2 - 1$ and $F(s) = s^{-\nu} G(\alpha_n s)$ reduces this equation to Bessel's equation of order ν. The solution regular at zero is a positive multiple of $s^{-\nu} J_\nu(\alpha_n s)$, with its value at zero understood by continuity. Positivity of F before its first boundary zero therefore gives $\alpha_n = j_{\nu,1}$, as stated in (5.5). In particular, $F > 0$ on $[0, 1)$. Integrating $(s^{n-1} F')' = -\alpha_n^2 s^{n-1} F$ from zero gives

$$F'(s) = -\alpha_n^2 s^{1-n} \int_0^s t^{n-1} F(t) \, \mathrm{d}t < 0 \qquad (0 < s \leq 1).$$

Set

$$V(z) = F(1 - z/L), \qquad 0 \leq z \leq L.$$

Then $V(0) = 0$, $V'(L) = 0$, and $V, V' > 0$ on $(0, L)$. Its equation is

$$-V''(z) + \frac{n-1}{L-z}V'(z) = \frac{\alpha_n^2}{L^2}V(z). \tag{5.7}$$

Since a differentiable concave function lies below each tangent line, $r(L) = 0$ implies

$$r(z) + (L-z)r'(z) \geq 0, \qquad \frac{r'(z)}{r(z)} \geq -\frac{1}{L-z} \quad (0 < z < L).$$

With $w(z) = r(z)^{n-1}$, equation (5.7) therefore yields

$$-\frac{(wV')'}{w} = -V'' - (n-1)\frac{r'}{r}V' \leq \frac{\alpha_n^2}{L^2}V. \tag{5.8}$$

This inequality is strict on an interval immediately to the right of 0: smoothness and reflection symmetry at the equator give $r'(0) = 0$, whereas $-1/(L-z)$ tends to $-1/L$.

Multiply (5.8) by wV and integrate. The boundary term wVV' vanishes at both endpoints: $V(0) = 0$, and $w(L) = 0$ while V, V' are bounded. Integration can first be performed on $[0, L-\varepsilon]$ and then passed to the limit; this justifies it also when the profile graph has an unbounded derivative at the pole. Indeed, w is decreasing, is locally absolutely continuous, and $\int_0^L |w'| \, \mathrm{d}z = w(0)$; hence the term $w'VV'$ is integrable. The other terms are integrable because V, V', V'' are bounded. The weighted deficit is positive on a fixed closed subinterval of the strictness region near zero. This interval is retained for all sufficiently small ε, so strictness survives the limit. The result is

$$\int_0^L w(V')^2 \, \mathrm{d}z < \frac{\alpha_n^2}{L^2}\int_0^L wV^2 \, \mathrm{d}z.$$

Oddly extend V to $[-L, L]$ and regard it as a function on Ω. Since $V(0) = 0$ and V' is bounded, this extension is Lipschitz and belongs to $H^1(\Omega)$. Fubini's theorem cancels the common factor $2\omega_{n-1}$ in its two Rayleigh integrals, giving exactly the preceding weighted quotient and proving the claim. □

Proposition 5.4 (Explicit aspect-ratio criteria)**.** *Either of the following conditions implies* (5.1)*:*

$$\frac{L}{R} \geq \frac{\alpha_n}{\beta_{n-1}}, \tag{5.9}$$

$$\frac{L}{R} \geq \sqrt{\frac{n+4}{2}}. \tag{5.10}$$

Proof. The exact condition (5.9), combined with Lemma 5.3, immediately gives (5.1). To obtain (5.10), test the Dirichlet quotient on B^n with $1 - |x|^2$. With $\omega_n = |B^n|$, its two integrals are

$$\int_{B^n} |\nabla(1-|x|^2)|^2 \, \mathrm{d}x = \frac{4n\omega_n}{n+2}, \qquad \int_{B^n} (1-|x|^2)^2 \, \mathrm{d}x = \frac{8\omega_n}{(n+2)(n+4)}.$$

The trial function is not a Dirichlet eigenfunction: its negative Laplacian is the constant $2n$, which cannot equal a positive eigenvalue times $1 - |x|^2$. Thus the variational inequality is strict, and

$$\alpha_n^2 < \frac{n(n+4)}{2}, \qquad \frac{\alpha_n^2}{\beta_{n-1}^2} < \frac{n+4}{2},$$

where the second inequality uses Lemma 5.1 with $m = n-1$. Thus (5.10) implies (5.9), in fact strictly. □

Remark 5.5 (A further elementary moment bound). Concavity also gives the explicit estimate

$$\frac{1}{|\Omega|}\int_\Omega z^2 \, \mathrm{d}x \geq \frac{2L^2}{(n+1)(n+2)}. \tag{5.11}$$

Indeed, $r(z)/(L-z)$ is nondecreasing on $(0,L)$ by the tangent-line inequality used above. Relative to the probability measure proportional to $(L-z)^{n-1}\,\mathrm{d}z$ on $(0,L)$, both $q(z)=(r(z)/(L-z))^{n-1}$ and z^2 are nondecreasing. Their covariance is nonnegative by the argument of Corollary 4.3. All the required moments are integrable because $q(z)(L-z)^{n-1}=r(z)^{n-1}$ and z^2 is bounded. It follows that

$$\frac{\int_0^L z^2 r(z)^{n-1}\,\mathrm{d}z}{\int_0^L r(z)^{n-1}\,\mathrm{d}z} \geq \frac{\int_0^L z^2(L-z)^{n-1}\,\mathrm{d}z}{\int_0^L (L-z)^{n-1}\,\mathrm{d}z} = \frac{2L^2}{(n+1)(n+2)}.$$

Consequently another sufficient condition is

$$\frac{L}{R} \geq \sqrt{\frac{(n+1)(n+2)}{2n}}. \tag{5.12}$$

Under this condition, (5.11) and $\beta_{n-1}^2 > n$ give

$$M_{\|} \geq \frac{2L^2}{(n+1)(n+2)} \geq \frac{R^2}{n} > \frac{R^2}{\beta_{n-1}^2}.$$

Proposition 5.2 therefore applies, including at the endpoint of (5.12). For $n>2$, (5.12) is slightly less restrictive than (5.10); at $n=2$ they coincide. Both avoid evaluating Bessel zeros.

5.4. **Completion of the main theorems.**

Proof of Theorems 1.1 and 1.2. Proposition 3.2 proves all spectral and multiplicity statements in Theorem 1.1. Propositions 3.3 and 4.1 prove its monotonicity and extremum statements, including the assertions on the two pure subspaces at a tie. Every linear combination within the transverse subspace is a positive multiple of u_e for a suitable unit vector e, by (3.4). Thus the transverse assertion applies to every first eigenfunction when that branch is strictly lower. The proved positive directional derivatives exclude interior critical points. An interior extremum would be a critical point, so both extrema on the compact closure occur on the boundary and every interior value lies strictly between them. This is (1.1).

For Theorem 1.2, the two axial conditions follow from Proposition 5.2 and Proposition 5.4, followed by Theorem 3.4. The elementary axial condition is (5.12). Its right-hand side is $\sqrt{10/3}<2$ at $n=3$ and $\sqrt{15/4}<2$ at $n=4$. The transverse conditions are Proposition 4.2 and Corollary 4.3. The strict trial inequalities in these arguments justify all endpoint cases. □

6. Balls, axial stretching, and examples with unbounded ratio

6.1. **The ball, including its full multiple eigenspace.**

Corollary 6.1. *Every nonzero first Neumann eigenfunction on B_R^n has a strictly positive derivative in some fixed direction. Its maximum and minimum on $\overline{B_R^n}$ are unique and occur at the two endpoints of the diameter in that direction. This statement includes every linear combination in the first eigenspace, whose dimension is n.*

Proof. The ball satisfies (G1)–(G3), and its orthogonal symmetries interchange the axial and transverse coordinates. Hence $\lambda_{\|}=\lambda_{\perp}$, and Proposition 3.2 gives $\dim E_1 = n$.

Given $0\neq u\in E_1$, rotate its nonzero vector $J(u)$ to a positive multiple of the axial unit vector. The rotated eigenfunction belongs to the axial line: Proposition 3.2 and injectivity of J eliminate all its transverse components. Proposition 3.3 applies at this tie and gives $J(U)\cdot e_z = \int_{B_R^n} U_z > 0$, so the axial coefficient is positive. Rotating its positive derivative and its two extrema back proves the assertion for u. □

6.2. **Only finitely many possible exceptional stretches.**

Proposition 6.2. *Fix a body Ω_0 satisfying* (G1)–(G3)*, with axial half-length L_0 and equatorial radius R_0. For $t > 0$, set*

$$\Omega_t = \{(y, tz) : (y, z) \in \Omega_0\}.$$

There is a finite nonempty set $\Sigma \subset (0, \infty)$ such that

$$\lambda_{\|}(\Omega_t) = \lambda_{\perp}(\Omega_t) \quad \Longleftrightarrow \quad t \in \Sigma.$$

For every $t \notin \Sigma$, every nonzero first eigenfunction on Ω_t has strict hot spots, unique maximum and minimum, and no interior critical point. The transverse conclusion holds for all sufficiently small t, and the axial conclusion for all sufficiently large t. No assertion about general mixed eigenfunctions at the values in Σ is made, apart from the ball case above.

Proof. The stretched profile is $r_t(z) = r_0(z/t)$ on $(-tL_0, tL_0)$; the stretching is an invertible linear map. It preserves the stated smoothness, symmetry, concavity, and strict profile decrease.

Pull functions back to the fixed domain Ω_0. The Jacobian factor t cancels from each Rayleigh quotient, leaving the form

$$q_t(f, g) = \int_{\Omega_0} (\nabla_y f \cdot \nabla_y g + t^{-2} f_z g_z). \tag{6.1}$$

In each of the two fixed coordinate-odd subspaces of $H^1(\Omega_0)$, its lowest eigenvalue is simple by Lemma 3.1, applied on Ω_t and pulled back by the same linear map. Both lowest eigenvalues are real analytic in $t > 0$.

Here is a form-level justification of this last assertion. Let V be one of those closed odd subspaces, equipped with its H^1 norm. The Poincaré inequality makes q_t coercive on V for each fixed $t > 0$, and the embedding of V into L^2 is compact. Let $A_t : V \to V^*$ denote the operator of the bilinear form (6.1), and let $B : V \to V^*$ denote the L^2 pairing. The map

$$(t, f, \lambda) \longmapsto \left(A_t f - \lambda B f, \ \frac{\|f\|_2^2 - 1}{2} \right) \quad \text{from } (0, \infty) \times V \times \mathbb{R} \text{ to } V^* \times \mathbb{R}$$

is analytic, since A_t depends analytically on t in operator norm. Fix $t_0 > 0$ and a normalized lowest eigenpair (f_0, λ_0) at t_0. The derivative in (f, λ) is

$$(h, s) \longmapsto ((A_{t_0} - \lambda_0 B)h - sBf_0, \ (f_0, h)_{L^2}).$$

This is an isomorphism. On the L^2-orthogonal complement of f_0, write $\lambda_+ > \lambda_0$ for the next eigenvalue in V. The min–max principle at t_0 gives

$$q_{t_0}(h, h) - \lambda_0 \|h\|_2^2 \geq \left(1 - \frac{\lambda_0}{\lambda_+}\right) q_{t_0}(h, h).$$

Uniform ellipticity of q_{t_0} and the Poincaré inequality on the odd subspace make the right-hand side coercive in H^1. More explicitly, for a target $(\ell, b) \in V^* \times \mathbb{R}$, write $h = bf_0 + h_\perp$ with $(h_\perp, f_0)_{L^2} = 0$. Testing the first component against f_0 forces $s = -\ell(f_0)$. The Lax–Milgram theorem on this closed orthogonal complement gives a unique $h_\perp$ satisfying

$$q_{t_0}(h_\perp, \varphi) - \lambda_0 (h_\perp, \varphi)_{L^2} = \ell(\varphi) \qquad (\varphi \in V, \ (\varphi, f_0)_{L^2} = 0).$$

Together with the equation tested against f_0, this solves the full derivative equation. Coercivity gives the bounded inverse. The analytic implicit function theorem supplies an analytic local eigenpair. For all $f \in V$,

$$\min\{1, (t_0/t)^2\} \, q_{t_0}(f, f) \leq q_t(f, f) \leq \max\{1, (t_0/t)^2\} \, q_{t_0}(f, f).$$

As $t \to t_0$, these comparison factors tend to one; min–max therefore gives continuity of the ordered eigenvalues and preserves the gap around λ_0. This identifies the local eigenvalue with the lowest

odd eigenvalue. The local eigenvalue branches agree on overlaps, proving analyticity of the two eigenvalues throughout $(0,\infty)$.

Let $M_{\|,0}>0$ and $M_{\perp,0}>0$ be the two second moments of Ω_0. Change of variables gives

$$M_{\|}(\Omega_t)=t^2M_{\|,0}, \qquad M_{\perp}(\Omega_t)=M_{\perp,0}.$$

The axial half-length is tL_0, while the equatorial radius remains R_0. For small t the transverse moment condition $M_{\perp,0}\geq 4t^2L_0^2/\pi^2$ holds, so Proposition 4.2 selects the transverse branch. For large t the axial condition $t^2M_{\|,0}\geq R_0^2/\beta_{n-1}^2$ holds, so Proposition 5.2 and Theorem 3.4 select the axial branch. Consequently the analytic function $\lambda_{\|}(\Omega_t)-\lambda_{\perp}(\Omega_t)$ is positive for all sufficiently small t and negative for all sufficiently large t. Its zero set is nonempty by continuity and is contained in a compact subinterval of $(0,\infty)$. A nonzero real analytic function has isolated zeros; infinitely many zeros in that compact interval would have an accumulation point. Hence the zero set Σ is finite. Theorem 1.1 gives all claimed eigenfunction conclusions off Σ. □

After pullback to Ω_0, every first eigenfunction with $t\notin\Sigma$ is either odd or even in z. These two parity subspaces are closed in $L^2(\Omega_0)$, as is their union. A mixture at a tie with nonzero axial and transverse parts belongs to neither subspace, so it cannot be an L^2 limit of these pure first modes. Thus the off-tie conclusions alone do not imply a hot-spots assertion for such mixtures.

Corollary 6.3. *For every fixed Ω_0 as in Proposition 6.2, all sufficiently large axial stretches have the strict hot-spots property, while*

$$\frac{d(\Omega_t)^n}{|\Omega_t|}\longrightarrow\infty \qquad (t\longrightarrow\infty).$$

Proof. The volume is $|\Omega_t|=t|\Omega_0|$, and the two axial poles are distance $2tL_0$ apart. Therefore

$$\frac{d(\Omega_t)^n}{|\Omega_t|}\geq\frac{2^nL_0^n}{|\Omega_0|}\,t^{n-1}\longrightarrow\infty.$$

The axial conclusion for large t was proved above. □

6.3. **An explicit smooth nonellipsoidal family.**

Proposition 6.4 (Quartic bodies)**.** *Fix $n\geq 2$, $R>0$, and $0<c<1$. For every $L>0$, the domains (1.8) are nonellipsoidal and satisfy* (G1)–(G3) *with strictly positive principal curvatures. They satisfy all axial hot-spots conclusions whenever*

$$\frac{L}{R}\geq\sqrt{\frac{n+2}{n}}. \tag{6.2}$$

Their volumes are

$$|\Omega_{L,R,c}|=\omega_{n-1}LR^{n-1}I_{n,c}, \qquad I_{n,c}=\int_{-1}^{1}[(1-s^2)(1+cs^2)]^{(n-1)/2}\,\mathrm{d}s. \tag{6.3}$$

If $L/R\geq\sqrt{1+c}$, their diameter is $2L$ and their exact diameter–volume ratio is (1.9). In particular, every sufficiently large L has the strict hot-spots property and the ratio tends to infinity as $L\to\infty$.

Proof. Let

$$F(y,z)=\frac{|y|^2}{R^2}+(1-c)\frac{z^2}{L^2}+c\frac{z^4}{L^4}.$$

Its Hessian is positive definite:

$$D^2F=\operatorname{diag}\left(\frac{2}{R^2},\dots,\frac{2}{R^2},\frac{2(1-c)}{L^2}+\frac{12cz^2}{L^4}\right).$$

Its gradient vanishes only at the origin, where $F = 0$, and $F(y,z) \geq |y|^2/R^2 + (1-c)z^2/L^2$ tends to infinity as $|(y,z)| \to \infty$. Hence the level $F = 1$ is a smooth compact hypersurface and bounds the convex domain $\{F < 1\}$. Its outward normal is $\nabla F/|\nabla F|$, so for a tangent vector T on this level,

$$\mathrm{II}(T,T) = \frac{D^2F(T,T)}{|\nabla F|} > 0 \qquad (T \neq 0),$$

which proves positivity of all principal curvatures.

Writing $s = z/L$, its profile is

$$r(z) = R\sqrt{1-(1-c)s^2-cs^4} = R\sqrt{(1-s^2)(1+cs^2)}. \tag{6.4}$$

Write $A(s) = 1-(1-c)s^2-cs^4$. It is positive on $(-1,1)$, $A'(s) < 0$ for $0 < s < 1$, and $A''(s) = -2(1-c) - 12cs^2 < 0$ on $(-1,1)$. Since

$$(\sqrt{A})'' = \frac{A''}{2\sqrt{A}} - \frac{(A')^2}{4A^{3/2}} < 0,$$

the profile satisfies $r'(z) < 0$ for $z > 0$ and $r''(z) < 0$ in $(-L, L)$. This verifies (G1)–(G3) explicitly, including the allowed square-root profile at each pole.

These domains are not ellipsoids. The unique center of an ellipsoid with these coordinate reflections would be fixed by each reflection, hence would be the origin. Those reflections eliminate all mixed quadratic terms, and transverse orthogonal symmetry makes the transverse coefficients equal. Its equation would therefore be $a|y|^2 + bz^2 < 1$, so its squared profile would be affine in z^2. The nonzero quartic coefficient in (6.4) makes that impossible.

Fubini and the substitution $z = Ls$ give (6.3). The axial second moment is

$$M_\parallel = L^2 \frac{\int_{-1}^1 s^2(1-s^2)^{(n-1)/2}(1+cs^2)^{(n-1)/2}\,\mathrm{d}s}{\int_{-1}^1 (1-s^2)^{(n-1)/2}(1+cs^2)^{(n-1)/2}\,\mathrm{d}s}.$$

With respect to the probability density proportional to $(1-s^2)^{(n-1)/2}$, the functions s^2 and $(1+cs^2)^{(n-1)/2}$ have strictly positive covariance by the two-point identity used in Corollary 4.3. Indeed, the second function is a strictly increasing function of s^2 because $c > 0$ and $n \geq 2$, and pairs with different values of s^2 have positive product measure. The untilted second moment is $1/(n+2)$, since

$$0 = \int_{-1}^1 \left[s(1-s^2)^{(n+1)/2}\right]' \mathrm{d}s = \int_{-1}^1 (1-(n+2)s^2)(1-s^2)^{(n-1)/2}\,\mathrm{d}s.$$

Consequently

$$M_\parallel > \frac{L^2}{n+2}. \tag{6.5}$$

Under (6.2), Lemma 5.1 therefore gives

$$M_\parallel > \frac{L^2}{n+2} \geq \frac{R^2}{n} > \frac{R^2}{\beta_{n-1}^2}.$$

The axial moment criterion applies even at the stated endpoint.

Finally, apply the diameter calculation to the centrally symmetric compact set $\overline{\Omega_{L,R,c}}$, which has the same diameter as $\Omega_{L,R,c}$. Its diameter is twice the maximum Euclidean norm: the triangle inequality proves the upper bound and an antipodal pair attains it. The maximum norm occurs on the boundary, since moving outward along a ray increases the norm. There, with $q = s^2 \in [0,1]$,

$$|(y,z)|^2 = R^2 + (L^2 - (1-c)R^2)q - cR^2q^2.$$

The derivative of this polynomial is bounded below by $L^2-(1+c)R^2$. If $L/R \geq \sqrt{1+c}$, it is therefore nonnegative on $[0,1]$, and the maximum is its value L^2 at $q=1$. Thus $d(\Omega_{L,R,c})=2L$. Combining this with (6.3) proves the exact ratio and its divergence. □

Corollary 6.5 (A single explicit family in every dimension)**.** *For every $n \geq 2$, $R>0$, and $L/R \geq \sqrt{2}$, the body*

$$\left\{(y,z) : \frac{|y|^2}{R^2}+\frac{z^2}{2L^2}+\frac{z^4}{2L^4}<1\right\}$$

has all the axial conclusions above, has diameter $2L$, and satisfies

$$\frac{d(\Omega)^n}{|\Omega|}=\frac{2^n}{\omega_{n-1}I_{n,1/2}}\left(\frac{L}{R}\right)^{n-1}.$$

Proof. Set $c=1/2$ in Proposition 6.4. For $n \geq 2$, one has $(n+2)/n \leq 2$ and $1+\frac{1}{2}<2$. The two conditions in Proposition 6.4 therefore hold. □

6.4. **Exact constants in dimension three.** For $n=3$ and $c=1/2$, the relevant integrals are elementary:

$$I_{3,1/2}=\int_{-1}^{1}\left(1-\frac{s^2}{2}-\frac{s^4}{2}\right)\,\mathrm{d}s=\frac{22}{15}, \qquad \int_{-1}^{1}s^2\left(1-\frac{s^2}{2}-\frac{s^4}{2}\right)\,\mathrm{d}s=\frac{34}{105}.$$

Thus the smooth positively curved nonellipsoidal domain

$$\Omega_{L,R,1/2}=\left\{(y_1,y_2,z) : \frac{y_1^2+y_2^2}{R^2}+\frac{z^2}{2L^2}+\frac{z^4}{2L^4}<1\right\}$$

has

$$|\Omega_{L,R,1/2}|=\frac{22\pi}{15}LR^2, \qquad M_{\|}=\frac{17}{77}L^2. \tag{6.6}$$

Already the elementary condition $L/R \geq 5/4$ guarantees both axial spectral selection and diameter $2L$. In fact,

$$\frac{17}{77}\left(\frac{5}{4}\right)^2-\frac{1}{3}=\frac{43}{3696}>0, \qquad \left(\frac{5}{4}\right)^2-\frac{3}{2}=\frac{1}{16}>0.$$

The first inequality and $\beta_2^2>3$ imply the axial moment criterion; the second implies the exact diameter condition. For every $L/R \geq 5/4$, therefore, every first eigenfunction has its unique extrema at the axial poles and

$$\boxed{\frac{d(\Omega_{L,R,1/2})^3}{|\Omega_{L,R,1/2}|}=\frac{60}{11\pi}\left(\frac{L}{R}\right)^2 \longrightarrow \infty.} \tag{6.7}$$

This family requires no smallness of a perturbation of the sphere.

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